

Detecting Liquid Food Adulteration with Mobile Hyperspectral Analysis based on RGB Camera

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Abstract

Liquid adulteration poses critical societal challenges by endangering public health and causing economic losses. Existing methods on mobile devices rely solely on vibration-based solutions, which face practical limitations due to requirements of specific container materials/structures and vibrational contact. In this paper, we present LiquidLens, a practical, non-contact mobile system for detecting liquid adulteration based on RGB images of liquid surfaces. Inspired by hyperspectral reconstruction, we aim to analyze liquid food's hyperspectral properties from RGB data for effective adulteration detection. To achieve this, we address three major challenges: First, the specular reflections of liquid, which act as a mirror image of surrounding obstacles, distort spectral measurements. Second, the camera's spectral range is restricted to the visible domain, which lacks access to vital information in the invisible bands. Third, the target liquid spectrum is entangled with uncertain ambient illumination, rendering it uninterpretable. To overcome these challenges, we first exploit the camera flash as visible light source and validate its spectral consistency across mobile devices. Next, we design a specular noise suppression technique to cancel the specular noise. Then, we design an RGB-based VIS-NIR spectral recovery (RVN-SR) framework to overcome the spectral range limitation. Besides, a cross-spectral training scheme is designed to train the RVN-SR model in the flash domain, compensating for the lack of NIR radiation. We deploy and validate LiquidLens across multiple smartphones. Experimental results show robust detection accuracy of 96.2% for 8 honey adulteration types and 94.0% for 5 wine adulteration types.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**.



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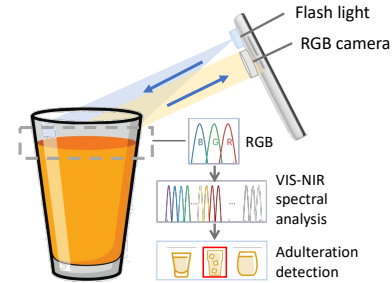


Figure 1: The concept of LiquidLens.

Keywords

Mobile Computing, Hyperspectral Reconstruction, VIS-NIR Spectroscopy, Liquid Food Adulteration Detection

1 Introduction

Liquid food adulteration is a critical societal issue, endangering public health and economic stability. Recent studies indicate that 46% of commercial honey products are adulterated [53], and 20% of wines are counterfeit [41]. These practices pose serious health risks: for example, adulterated honey has been linked to hypercholesterolemia and allergic reactions [19]. In more severe cases, as reported by the BBC, there have been numerous cases involving alcohol adulterated with toxic polyols—leading to poisoning and even fatalities [1, 10, 68]. Liquid adulteration also causes substantial economic losses, amounting to billions of dollars globally each year, solely due to wine and honey fraud [7, 46]. Investigations have revealed that regulatory oversights have allowed adulterated liquid food products to flood the market [3, 24]. Therefore, it is crucial to develop detection methods for liquid food adulteration that are readily accessible to consumers for daily use.

Mobile devices present a promising platform to achieve this goal, given their ubiquity and various sensors. Recently, a growing number of liquid detection studies have utilized various sensors on unmodified mobile devices, such as cameras [60, 74], IMUs [30, 70], and microphones [38]. These methods analyze how a liquid modulates vibrations to infer its viscosity, using either direct measurement (e.g., IMU or

microphone signals) or indirect observation (e.g., visually tracking surface ripples or bubble movements via camera). However, these vibration-based sensing methods require particular container materials or structures to function effectively, and requires direct vibrational contact or manual shaking, causing practical limitations.

In this paper, we therefore ask: can we realize a practical, non-contact liquid food adulteration detector using only a commercial mobile device? We shift the sensing paradigm from bulk viscosity to molecular specificity. Unlike prior methods, our approach leverages the phone’s camera for direct spectral analysis to target the unique molecular fingerprints of a liquid. The core principle is that variations in liquid food composition (such as those caused by adulteration) can be revealed through spectral analysis [31]. While prior studies have successfully utilized hyperspectral devices [29, 44, 47] to achieve precise liquid food adulteration detection, such equipment is typically bulky, expensive, and unsuitable for daily applications. In contrast, mobile devices are generally equipped with RGB cameras, which lack the spectral wavelength resolution required for detailed hyperspectral analysis. Fortunately, we are inspired by the hyperspectral reconstruction technology [28, 56, 75], which enables hyperspectral reconstruction in the visible range with higher wavelength resolution from conventional RGB inputs. This technology enables hyperspectral analysis based on mobile RGB data for liquid adulteration detection.

However, applying hyperspectral analysis to mobile RGB cameras for liquid adulteration detection is not a trivial task. It faces the following challenges: 1) **Complex Liquid Surface Reflection**: Liquid surface reflection includes two components: bulk reflection from light interacting with the liquid’s interior, providing spectral detection data; and specular reflection, a mirror-like effect from surrounding objects that distorts the spectrum. While conventional hyperspectral systems demand isolated environments to avoid the specular interference [9, 29, 37, 44], our solution functions in open settings where nearby object reflections can cause interference. Thus, extracting the bulk reflection signal from this mixed spectrum remains a key challenge. 2) **Visible-Limited Spectral Sensing Constraint**: Existing hyperspectral reconstruction methods are limited to acquiring fine-grained spectral resolution only within the visible range (400–700 nm) from RGB images captured by mobile devices. However, the near-infrared (NIR) region (>700 nm) provides essential complementary information for liquid food adulteration analysis [48, 65], as it captures molecular signatures that are undetectable within the visible band. Consequently, conventional mobile RGB sensors offer no measurable response in the NIR region, presenting a significant gap for comprehensive spectral analysis. Note that several flagship smartphones have begun to incorporate dedicated NIR imaging modules,

such implementations remain rare [55]. 3) **Uncertain Illumination Spectrum for Liquid**: The reflected spectrum of liquid exhibits significant dependence on illumination conditions. When measured under diverse ambient illumination, the spectral reflectance profile of identical liquid food samples exhibits variability. To address this challenge, conventional hyperspectral solutions rely on controlled illumination conditions, which are impractical for mobile device users.

We propose LiquidLens, a mobile system for liquid food adulteration detection that exclusively utilizes the RGB sensing capability of mobile devices to perform VIS-NIR spectral analysis, as shown in Fig. 1. We observe that the camera flash can serve as a spectrally consistent light source across mobile devices. Integrated as ordinary components in mobile devices, camera flashes provide visible-spectrum illumination and serve as essential supplemental light sources for low-light photography. We analyze the spectral diversity of common light sources and then investigate mobile camera flashes across multiple devices, demonstrating their remarkable spectral consistency (Section 4). To remove the specular reflection interference, we design a specular noise suppression technique. We first analyze the optical path at liquid surfaces carefully and find that the fixed camera-flash geometry allows a common viewing tilt to steer the flash’s own specular reflection out of view. This leaves only the ambient light’s specular component, which we suppress by computing a “flash-only” image (obtained by subtracting an ambient-only capture from a flash-illuminated one). This process effectively isolates the underlying bulk reflection (Section 5). To break the limited spectral sensing range, we propose a RGB-based VIS-NIR Spectral Recovery (RVN-SR) module to recover the NIR spectral information (Section 6.1). Here, we first explore the correlation between visible and NIR spectral bands. Our analysis reveals that this correlation does exist, but it is observed only sparsely. Thus, we develop a Sparsity-Gated NIR Transformer (SNT) structure that suppresses irrelevant spectral noise to recover the NIR spectral information. In addition, since camera flashes lack NIR spectral components, we design the expert-guided cross-spectral domain adaptation (ECDA) framework to train the RVN-SR module. ECDA solves the data limitation by transferring spectral correlations learned from full-spectrum sources (with complete VIS-NIR ground truth) to the flash domain, bridging the domain shift between broad-band and flash-specific illumination (Section 6.2).

We implement LiquidLens on commercial mobile devices and demonstrate its effectiveness in detecting adulteration in honey and wine. For honey, eight distinct types of adulterants are considered, including water, cheap honey, syrups, and illegal additives. For wine, five categories of adulterants are evaluated, such as water, cheap alcohol, fruit juice,

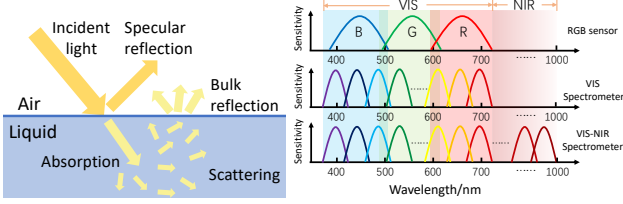


Figure 2: Specular reflection and bulk reflection.

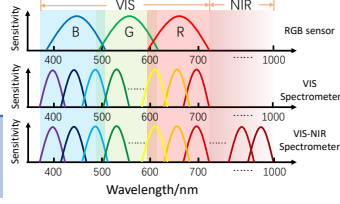


Figure 3: Sensitivity of RGB and hyperspectral device.

and illegal additives. Various concentration levels are tested for each adulterant. Experimental results show that LiquidLens achieves an accuracy of 96.2% in identifying 8 honey adulteration types and 94.0% for 5 wine adulteration types. Furthermore, we validate the robustness of LiquidLens under diverse environmental conditions, including 5 different illumination environments, 8 different capture distances, and across 3 different device models.

In summary, we make the following contributions:

- We developed LiquidLens, the first mobile-based system capable of detecting liquid food adulteration based on hyperspectral analysis. Leveraging only the ubiquitous RGB camera and flash standard on mobile devices, our system achieves accurate liquid authentication with the capability to identify specific adulterants.
- We present a novel framework for VIS-NIR spectral recovery by leveraging a mobile device’s RGB camera, which substantially improves the detection accuracy and reliability of liquid food adulteration.
- We implement and evaluate LiquidLens. Experimental results demonstrate LiquidLens’s high accuracy in detecting various adulteration types and robustness under different experimental settings.

2 Background and Challenges

2.1 Background

VIS-NIR Spectroscopy for Liquid Food Adulteration. VIS-NIR spectroscopy is a well-established technique for detecting liquid food adulteration, operating primarily within the 400–1000 nm spectral range [6, 20, 37, 52, 54, 73]. The core idea is to measure how the liquid food interacts with light, specifically its absorption quantified via reflectance. In the visible spectrum (400–700 nm), the electronic transitions of chromophores (e.g., flavonoids, carotenoids) directly reveal color and pigment concentration [4, 51]. Moving into the near-infrared (NIR) region (700–1000 nm), the technique captures overtone and combination vibrational bands, predominantly from O–H and C–H bonds, yielding valuable information on moisture content, sugar structure and lipid composition[15].

Liquid Surface Reflection. Light reflection from a liquid surface consists of two distinct components: specular reflection and bulk reflection (Fig. 2). Specular reflection, which is *directional* [66], occurs at the air-liquid interface and follows the law of reflection. It is independent of the liquid’s internal composition, as it involves no penetration into the medium. In contrast, bulk reflection, which is *non-directional*, arises from light that enters the liquid and interacts with its molecular constituents and suspended particles. This component carries spectral information related to chemical bonds (e.g., O–H, C–H, N–H) and scattering elements, making it highly dependent on the liquid’s composition.

Optical Imaging Principle. In an optical imaging system, each pixel converts incident light into an electrical signal via its spectral response. The pixel value I is the integral of the light intensity over the sensor’s wavelength range, weighted by three spectral components: the illumination spectrum $S(\lambda)$, the object reflectance $R(\lambda)$, and the sensor response $C(\lambda)$ [55]:

$$I \propto \int_{\lambda_{\min}}^{\lambda_{\max}} S(\lambda)R(\lambda)C(\lambda) d\lambda. \quad (1)$$

Thus, the output is a spectrally weighted sum determined by the interplay of these three factors.

Hyperspectral Reconstruction. Hyperspectral reconstruction [8, 56, 75] recovers spectral data with high wavelength resolution from standard RGB inputs. As illustrated in Fig. 3, RGB captures color via three broad bands, while hyperspectral devices (e.g. spectrometer) sample many narrow contiguous bands for fine spectral detail. The core principle is that each RGB value results from a spectrally-weighted integration of the incident light across the entire visible spectrum. Models trained on paired RGB and hyperspectral data learn the mapping from broad-band responses to underlying spectral features, thereby reconstructing continuous high-resolution spectra through learned correlation.

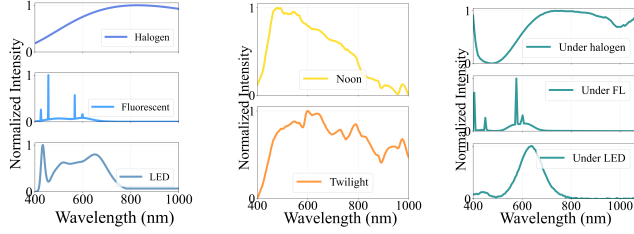
2.2 Challenges

Our goal is to realize liquid food adulteration detection based on liquid VIS-NIR spectral analysis using the built-in RGB image sensor on mobile devices. This task, however, faces several significant research challenges, summarized as follows:

Complex Liquid Surface Reflection. As discussed in Section 2.1, light reflected from a liquid surface contains both specular and bulk components. The bulk reflection carries the spectral fingerprint crucial for chemical characterization, yet remains mixed with specular reflection. As shown in Fig. 4, the specular interference, which visually appears as a mirror image of the surrounding scene, acts as a disruptive



Figure 4: Specular reflection of a clock and curtain at the liquid surface.



(a) Artificial lighting **(b) Nature lighting**

Figure 5: Spectra of different lighting.

Figure 6: The reflect spectrum under different illuminations.

noise source that can distort the spectral features embedded within the target bulk reflection. To achieve accurate results, traditional hyperspectral analysis of liquids is performed in isolated environments to control specular reflection [9, 29, 37, 44]. However, our method operates in open environments where nearby objects induce specular reflections. Therefore, the effective separation of this interfering specular reflection is essential to reliable spectral analysis of the target liquid.

Visible-Limited Spectral Sensing Constraint. As shown in Fig. 3, RGB cameras are limited to capturing data within the visible band and lack access to the NIR region. Specifically, RGB sensors acquire information only within the visible spectrum (approximately 400–700 nm), whereas VIS-NIR hyperspectral instruments (e.g., spectrometers) cover an extended range from 400 nm to 1000 nm with higher wavelength resolution, thereby incorporating chemically meaningful NIR spectral features. Although hyperspectral reconstruction techniques can enhance spectral wavelength resolution within the visible range using RGB data, they remain unable to recover the NIR portion (700–1000 nm) due to its complete absence in the original signal.

Uncertain Illumination Spectrum for Liquid. According to Eq. (1), the ambient light spectrum $S(\lambda)$ weights wavelengths differently, so the same reflectance $R(\lambda)$ yields different reflected intensities under distinct illuminants (Fig. 6). Traditional hyperspectral methods require known and controlled illumination, whereas mobile devices operate under varying conditions (e.g., artificial and natural lighting). Figs. 5a and 5b demonstrate substantial spectral differences across these conditions. Without a constant $S(\lambda)$, accurately

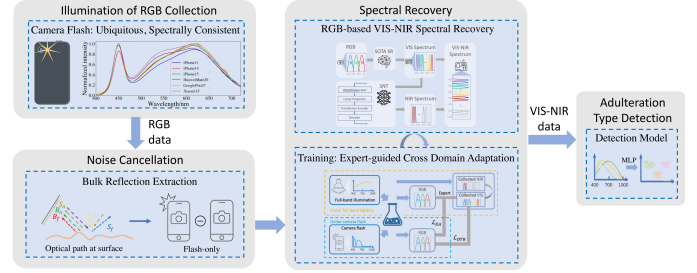


Figure 7: The system structure of LiquidLens.

extracting the target’s spectral signature $R(\lambda)$ becomes an ill-posed problem.

3 Overview

In this section, we provide the system overview of LiquidLens. As shown in Fig. 7, LiquidLens is mainly composed of three modules: First, we utilize the mobile camera flash as a spectrally consistent light source across mobile devices, as established in this work. We measure the spectral data of camera flashes across various mobile devices to demonstrate that mobile flashes exhibit inherent spectral consistency. Then, based on the analysis of the optical path at liquid surfaces, we utilize the co-location of flash light and camera and design a specular noise suppression technique. This scheme computes a flash-only image (derived by subtracting ambient-only images from flash-illuminated captures) to effectively remove specular noise and isolate the bulk reflection signature. Finally, we design an RGB-based VIS-NIR Spectral Recovery (RVN-SR) module to recover the missing NIR information. It is a transformer-based architecture which leverages our insights into the sparse relationship between visible and NIR spectral bands. Furthermore, since camera flashes lack NIR components, training requires full-band illumination. However, the spectral discrepancy between light sources introduces domain shift. To address this, we introduce the Expert-guided Cross-spectral Domain Adaptation (ECDA) framework, which uses NIR data from full-band illumination to train RVN-SR and bridge the domain gap across lighting conditions.

4 Camera Flash: A Spectrally Consistent Light Source across Mobile Devices

According to the analysis in Section 2.2, the spectral variability in ambient light compromises the reliability of liquid spectral analysis under uncontrolled lighting conditions. While lab spectrometers provide controlled, standardized illumination, mobile environments present uncontrolled and spectrally variable conditions. Here, we first analyze common illumination spectra, sourcing artificial light data from

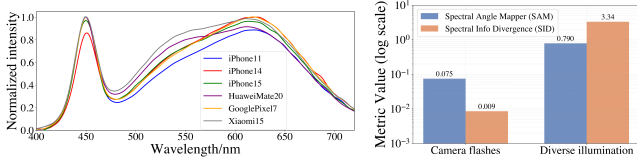


Figure 8: Camera flash spectra on different mobile devices. **Figure 9: SAM and SID for camera flashes and diverse illumination.**

CIE [14] and generating natural light spectra via SMARTS [25]. Fig. 5a and 5b show their spectral distributions. For artificial illumination, halogen bulbs emit a broad and uniform spectrum, while LEDs and fluorescent lamps are limited to the visible band, with fluorescent spectra showing sharp peaks and LEDs having distinct spectral shapes. For natural illumination, sunlight varies throughout the day, shifting toward longer wavelengths at twilight. Thus, it is difficult to find a consistent and continuous illumination environment in real-world conditions.

Fortunately, integrated smartphone camera flashes provide a spectrally consistent visible illumination source across mobile devices. Most mobile flashes utilize LEDs, offering advantages in compactness, power efficiency, and longevity [35]. Crucially, unlike variable ambient light, flash spectra remain consistent across smartphone models due to standardized LED materials and fabrication [50, 57, 64]. In order to confirm the spectral consistency of camera flashes on mobile devices, we measure flash spectra across 6 device models (i.e., iPhone 11, iPhone 14, iPhone 15, HuaweiMate 20, Google Pixel 7, Xiaomi 15) using a spectral illuminance meter (AH-310). Results are shown in Fig. 8. We can see that the spectral profiles of mobile flashes across various devices consistently display a sharp emission peak centered at 435–470 nm, accompanied by a broader secondary peak covering the 500–700 nm range. To quantify spectral differences and LED consistency, we employed Spectral Angle Mapper (SAM) [69] and Spectral Information Divergence (SID) [55]. SAM measures angular similarity, while SID assesses brightness distribution divergence. We then compared mean SAM and SID values across diverse light sources and smartphone flashlights. As shown in Fig. 9, the results demonstrate significantly lower SAM and SID values for camera flashes compared to conventional light sources across all categories, which confirms the spectral consistency of camera flashes.

5 Bulk Reflection Extraction

In this section, our goal is to extract the bulk scattering component from flash-illuminated images captured by mobile devices. To this end, we suppress non-informative specular reflections by using the flash-only image (the difference

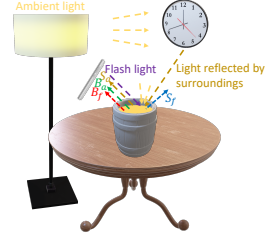


Figure 10: Optical path analysis at the liquid interface.

between flash-illuminated and ambient-only images) as the basis for bulk scattering extraction. Our key insight leverages a fundamental advantage of the smartphone platform: the co-location of flash light and camera on the same device. Because they are fixed together, any angular motion of the phone is shared by both units. As shown in Fig. 10, by applying a suitable tilt, this shared movement actively steers the directional specular highlight from camera flash (S_f) away from the camera’s field of view, while the non-directional bulk scattering from the liquid volume (B_f) continues to be captured. Under this configuration, the contribution of S_f becomes negligible.

Formally, a flash-illuminated image (I) and an ambient-only image (I_a) can be expressed as: $I = S_a + B_a + S_f + B_f$, $I_a = S_a + B_a$, where S_a and B_a are the specular and bulk reflected components from ambient light, while S_f and B_f are those from camera flash. Note that the specular components S_a and S_f do not encode chemical information about the liquid, despite governing its surface appearance.

We first use feature-based alignment to align this with the flash-illuminated image and then perform subtraction, which yields: $I_{fo} = I - I_a = (S_a + B_a + S_f + B_f) - (S_a + B_a) = S_f + B_f$. Although the derivation retains the specular component S_f , we now demonstrate its negligible contribution. As shown in Fig. 10, due to the coordinated tilt maneuver described above, the flash-induced specular reflection is effectively excluded from capture. Furthermore, any secondary specular reflection of the flash from environmental objects is severely attenuated and considered negligible. Thus, $S_f \approx 0$. Therefore, I_{fo} effectively isolates the bulk scattering component from the flash illumination: $I_{fo} \approx B_f$.

Thus, I_{fo} isolates the chemically informative bulk reflection from the original surface reflection, providing a reliable input for subsequent chemical analysis.

6 From RGB to Detection: An Integrated Spectral Recovery Framework

Currently, we have acquired bulk reflection RGB images under consistent illumination conditions. For each image, a Region of Interest (ROI) is manually defined to isolate the liquid sample, within which each pixel is treated as an

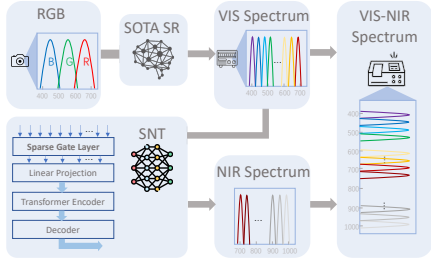


Figure 11: The RGB-based VIS-NIR Spectral Recovery (RVN-SR) model.

independent observation. Next, we perform VIS-NIR spectral analysis based on the RGB value of a single pixel from the image.

However, two significant challenges arise: First, due to inherent limitations of RGB image sensors, the spectral resolution and range of RGB data are substantially lower than those of VIS-NIR spectral data. Second, the camera flash illumination spectrum does not cover the NIR band, resulting in the absence of reflected NIR spectrum data from the target liquid under camera flash.

In this section, we first introduce the RGB-based VIS-NIR Spectral Recovery (RVN-SR) model designed to simultaneously enhance spectral resolution and extend the spectral range of RGB sensor. Note that the model assumes full-band VIS-NIR illumination conditions, with corresponding ground truth spectra available for model training. Subsequently, we propose an Expert-guided Cross-spectral Domain Adaptation (ECDA) training scheme to address hardware limitations of mobile flashlights, which lack NIR spectrum coverage. Finally, we introduce the adulteration detection model based on the spectral analysis result.

6.1 VIS-NIR Spectral Recovery Under Full-band Illumination

In this section, we introduce the RVN-SR design. Our goal is to recover the VIS-NIR spectrum based on RGB data.

6.1.1 From RGB to VIS Spectrum. Deriving visible spectrum from RGB data is inherently a task of spectral super-resolution, also referred to as hyperspectral reconstruction, which is a well-established topic [8, 56, 75]. Here, we leverage a state-of-the-art (SOTA) spectral reconstruction algorithm [69] to achieve high-wavelength-resolution spectral estimation $\hat{Spec}$ from RGB inputs.

To adapt this architecture for our task, we implement a focused modification. The original design in [69] employs a dual-branch structure with both spectral and spatial modeling modules. Since our emphasis lies on chemical composition analysis rather than spatial detail recovery, we retain

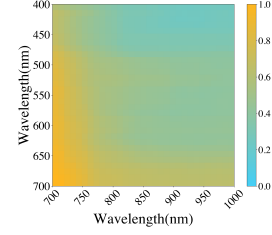


Figure 12: NMI across VIS-NIR wavelengths.

only the spectral attention block and remove the spatial branch. Additionally, we streamline loss computation by replacing pixel-wise metrics with a wavelength-aggregated measure. The loss is defined as the mean relative absolute error (MRAE) [69] across all reconstructed visible spectral bands: $\mathcal{L}_{SR} = \frac{1}{n_{vis}} \sum_{i=1}^{n_{vis}} \frac{|Spec(i) - \hat{Spec}(i)|}{|Spec(i)|}$, where n_{vis} is the total number of visible sampled wavelengths, $Spec(i)$ represents the ground-truth of visible spectral value at the i -th wavelength, and $\hat{Spec}(i)$ denotes the reconstructed spectral value. By minimizing the loss $\mathcal{L}_{SR}$, the model can narrow the distance between the reconstructed and original visible spectra.

6.1.2 From VIS to NIR Spectrum. Following the visible spectral reconstruction, we obtain a high-resolution visible spectrum from RGB data. However, directly predicting the NIR spectrum from RGB inputs remains infeasible, as RGB signals inherently lack NIR spectral information. Although RGB values only represent integrals of the visible spectrum, we identified intrinsic correlations between the reconstructed visible spectrum and the NIR bands. Thus, as shown in Fig. 11, instead of directly predicting NIR spectra from RGB inputs, we utilize the reconstructed visible data to infer the NIR spectrum. In this section, we first analyze correlations between the visible and NIR bands, then introduce the NIR predictor model.

VIS-NIR Cross-Spectral Correlations. The VIS-NIR cross-spectral correlations in liquid foods arises because multiple constituents vary together and their spectral effects add up. In the visible range, signals are largely shaped by chromophores (for example, pigments, or phenolic compounds), while the NIR is influenced by overtone/combination bands of abundant O-H/C-H groups in major constituents (for example, water, sugars, alcohols, or acids). These components tend to co-vary with source, formulation, and processing, so shifts in composition move both VIS and NIR in tandem, yielding measurable mutual information. Here we calculate the Normalized Mutual Information (NMI) across VIS-NIR

wavelengths to quantitatively validate cross-spectral dependencies. NMI quantifies the normalized statistical dependence between two variables by measuring shared information relative to their individual entropies (H), capturing both linear and nonlinear correlations. In contrast to Mutual Information (MI), NMI scales this value to $[0,1]$, allowing consistent comparison across variables with different marginal distributions. The NMI between visible band V_i and NIR band W_j is [34]:

$$\begin{aligned} \text{NMI}(V_i; W_j) &= \frac{2 \cdot \text{MI}(V_i; W_j)}{H(V_i) + H(W_j)} \\ &= \frac{2 \cdot \sum_{v \in V_i} \sum_{w \in W_j} p(v, w) \log_2 \frac{p(v, w)}{p(v)p(w)}}{- \sum_{v \in V_i} p(v) \log_2 p(v) - \sum_{w \in W_j} p(w) \log_2 p(w)}, \end{aligned} \quad (2)$$

where V_i and W_j denote the value ranges of visible band i and NIR band j , respectively. Here, v and w represent reflectance values within each band; $p(v)$, $p(w)$ are their marginal probability distributions, and $p(v, w)$ is the joint distribution.

Here we analyze the honey hyperspectral dataset [49] as an example. As shown in Fig. 12, adjacent spectral bands exhibit stronger correlations, peaking near 700 nm and decreasing toward both 400 nm and 1000 nm. This occurs because materials interact with light over continuous spectral ranges rather than at isolated wavelengths, causing neighboring bands within the same spectral feature to capture similar interactions. Moreover, each NIR band shows significant correlation ($\text{NMI} > 0.5$) with certain visible bands, confirming their inherent link and supporting visible-based NIR prediction [16]. However, we notice that the correlation between visible and NIR bands is sparse. Each NIR band exhibiting correlations with only a small subset of visible bands.

NIR Predictor Model. As we’ve analyzed above, Fig. 12 demonstrates that sparse correlations exist between visible and NIR bands. Consequently, leveraging full visible spectral information for NIR prediction introduces redundant data that increases computational burden and propagates noise interference. We propose the Sparse-Gated NIR Transformer (SNT), an architecture that leverages sparsity-optimized transformer operations on reconstructed visible spectra for NIR prediction. The model initiates processing with a sparse, trainable gating layer, which applies sigmoid-activated weights to dynamically identify and enhance a sparse subset of task-relevant wavelengths while suppressing irrelevant or noisy bands. This sparsity-inducing gating mechanism ensures that only a limited number of spectral features contribute significantly to downstream processing. The processed input is projected to 64 dimensions via a linear layer, then encoded by a 3-layer Transformer (4

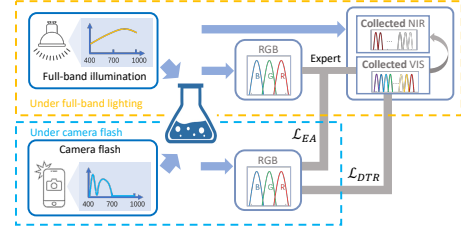


Figure 13: Expert-guided Cross-spectral Domain Adaptation (ECDA) training framework.

heads/layer) with self-attention and GeLU-activated feedforward networks. Features are decoded through a 128-neuron ReLU linear layer and projected to NIR wavelengths. The model’s loss is the MRAE computed across all NIR spectral bands: $\mathcal{L}_{SNT} = \frac{1}{n_{nir}} \sum_{i=1}^{n_{nir}} \frac{|Spec(i) - \hat{Spec}(i)|}{|Spec(i)|}$, where n_{nir} is the total number of NIR sampled wavelengths, $Spec(i)$ represents the ground-truth of NIR spectral value at the i -th wavelength, and $\hat{Spec}(i)$ denotes the recovered NIR spectral value. Minimizing $\mathcal{L}_{SNT}$ reduces the discrepancy between recovered and original NIR spectra.

6.2 Spectral Recovery Training Scheme Under Camera Flash

As shown in Fig. 8, camera flashes inherently lack NIR spectral components, making it infeasible to obtain ground-truth NIR data under camera flash illumination. One simple solution is to collect paired VIS-NIR ground truth data using a full-band illumination source, which emits a substantially full VIS-NIR spectrum. However, directly applying a model trained on such data to flash conditions results in a significant domain shift due to the substantial difference in spectral distribution between full-band and flash illumination, even within their shared visible spectral band, as shown in Fig. 5a.

To overcome these limitations and effectively leverage the available expert knowledge, we propose the Expert-guided Cross-spectral Domain Adaptation (ECDA) framework. As shown in Fig. 13, ECDA uses a two-stage teacher-student strategy. First, a teacher model is trained under full-band illumination to reconstruct visible spectra, learning robust spectral representations. Its parameters are then frozen, and intermediate features f_t are extracted for subsequent knowledge transfer. Next, a student model is trained on flash RGB images. During this student training phase, we extract the corresponding features from the student, denoted as f_s , and perform feature-level alignment by minimizing the representation gap between f_t and f_s through feature distillation. Note that the visible spectral reconstruction network in Section 6.1.1 uses a multistage architecture with spectral attention blocks. Here, the feature alignment is applied to the

first-stage outputs to transfer spectral knowledge from the teacher to the student. This alignment loss is combined with the spectral reconstruction loss to form the overall training objective. Specifically, the loss function is:

$$\mathcal{L}_{ECDA} = \lambda \cdot \mathcal{L}_{DTR} + (1 - \lambda) \cdot \mathcal{L}_{EA}, \quad (3)$$

where $\mathcal{L}_{DTR}$ is the domain-transfer regression loss, which utilizes full-band ground truth to directly supervise the flash model’s spectral reconstruction. $\mathcal{L}_{EA}$ represents expert alignment loss, which aligns intermediate features between the flash and expert models to reduce cross-domain discrepancy. The coefficient λ is a trade-off hyperparameter that balances the contributions of the two loss terms, which is 0.7 in our method. $\mathcal{L}_{DTR}$ and $\mathcal{L}_{EA}$ are represented as: $\mathcal{L}_{DTR} = \frac{1}{n_{vis}} \sum_{i=1}^{n_{vis}} \frac{|f_{full-band}(i) - f_{flash}(i)|}{|f_{full-band}(i)|}$, $\mathcal{L}_{EA} = \frac{1}{n_f} \|f_t - f_s\|^2$. Here n_{vis} is the total number of sampled visible wavelengths, $f_{full-band}(i)$ represent the original visible spectral values under full-band illumination at the i -th wavelength, respectively; $f_{flash}(i)$ denotes the reconstructed visible spectral value under camera flash. n_f is the dimension of feature alignment, which is set to 64 in our implementation. This mechanism enables accurate visible spectral reconstruction under camera flash illumination despite the absence of target-domain supervision. Following the cross-domain adaptation from flash to full-band visible spectra, the resulting outputs are used with full-band NIR data to train the NIR predictor.

6.3 Adulteration Detection Model

After recovering the hyperspectral data from RGB inputs via the model above, we obtain hyperspectral information with dimensions $n_{vis} + n_{nir}$, spanning both visible and NIR bands. This high-dimensional spectral information serves as the foundational input for classifying liquid food adulteration types. To achieve this, we employ a Multilayer Perceptron (MLP) classifier comprising three fully connected hidden layers (250, 150, and 100 neurons respectively) to detect adulteration patterns effectively.

7 Evaluation

7.1 Dataset for Spectral Recovery

We employ a Cubert UHD 185 hyperspectral camera to acquire VIS-NIR hyperspectral ground truth data of liquid food samples. The hyperspectral images are collected under halogen illumination for various types of liquid foods, as well as for samples adulterated with different contaminants listed in Table 1 and Table 2. Note that for adulterated samples, hyperspectral data acquisition is limited to the maximum concentration of each adulterant in Table 2. Two samples are prepared for both the authentic liquid food and the adulterated one. Thus we have 18 (2 honey types $\times$ 8 adulterant

Table 1: Experimental Settings.

Item	Number	Value
Liquid food	2	Honey: Manuka, Acacia Wine: Merlot, Cabernet
Adulterant	8	Shown in Table 2
Environment	5	Dark room, conference room, office, home, beside a window
Distance	8	5 cm, 7 cm, 10 cm, 20 cm, 30 cm, 40 cm, 50 cm, 55 cm
Device	3	iPhone 11, iPhone 15, Redmi note 11
Bottle	2	Seal, open

Table 2: Adulterants Types and Concentrations.

	Type	Substance	Conc.
Honey	Water	Water	20%, 40%
	Substitute	Multi-flower, rapeseed honey	20%, 50%
	Syrup	Maltose, invert, corn syrup	20%, 40%
	Illegal additive	Tartrazine Glycol	0.1, 0.2g/kg 2%, 5%
Wine	Water	Water	20%, 40%
	Substitute	Brandy	20%, 50%
	Juice	Grape juice	20%, 40%
	Illegal additive	Carmin Glycerol	0.1, 0.2g/kg 5%, 10%

types + 2 authentic types) $\times$ 2 = 36 honey samples and 12 (2 wine types $\times$ 5 adulterant types + 2 authentic types) $\times$ 2 = 24 wine samples in total. Thus, the complete dataset consists of 6,000 spectral data points.

Our spectral recovery model, as described in Section 6, requires three key components for training: (1) RGB data captured under full-band illumination, (2) RGB data under camera flash illumination, and (3) hyperspectral data acquired under full-band illumination. While our current dataset only contains hyperspectral measurements obtained under full-band illumination, we can derive the necessary RGB values through color conversion methodologies. Full-band RGB values are computed directly from the hyperspectral data using the CIE 1931 color-matching functions [13], as RGB is fundamentally derived from spectral integration across visible wavelengths. We employ an illumination conversion method [61] to transform the RGB values of samples from halogen to flash illumination. This leverages the principle that the reflected light from an object represents the coupled interaction between the illumination spectrum and the object’s reflectance spectrum [18]. We first measure the spectral data

of the halogen light source. Combined with the known spectral data of the smartphone flash, this allows us to compute the corresponding RGB values under flash illumination based on the measured halogen-illuminated RGB values. Here, we use the spectrum of iPhone 11 camera flash measured in Section 4.

7.2 Dataset for Liquid Food Adulteration Detection

To evaluate the detection accuracy of LiquidLens for liquid food adulteration types, we assemble a dataset with RGB cameras from different mobile device models.

Experimental setup. We collect images of 2 types of liquid food: honey and wine. Each type of liquid food, we collect images of 2 types. For honey, we collect Manuka honey (\$18.1 per 500g) and acacia honey (\$7.7 per 500g). For wine, we select Merlot from the Languedoc region of France (\$10.1 per 500mL), and Cabernet Sauvignon from California (\$15.4 per 500mL). For honey, we examine four categories of adulterants: water, substitutes (low-cost honey), syrup, and illegal additives. Note that the cheap honey we use here is priced below \$3.5 per 500g. From each category, several representative adulterants are selected, yielding a total of five distinct adulterants per honey type. For wine, we also consider four types of adulterants: water, substitutes (low-priced grape brandy), grape juice, and illegal additives. Here, the grape brandy cost less than \$5 per 500mL. Multiple typical adulterants are chosen from each category, resulting in five different adulterants per wine type. We evaluate 2 typical adulteration concentrations [5, 26, 32, 59] for each adulterant. Note that the concentration values represent mass fractions. Thus, we have 34 types (2 honey types $\times$ 8 adulterant types $\times$ 2 concentration types + 2 authentic types) of honey samples and 22 types (2 wine types $\times$ 5 adulterant types $\times$ 2 concentration types + 2 authentic types) in total. We evaluate LiquidLens in 5 different environments, at 8 different distances, using 3 different mobile device models, and under two distinct conditions: with the bottle cap open (capturing the liquid surface directly) and with the bottle cap closed (capturing the liquid through the bottle). Table 1 and Table 2 illustrates our experimental settings.

Liquid sample preparation. We prepare liquid samples for two types of honey, two types of wine, and their adulterants, with each adulterant prepared at two representative concentration levels [5, 26, 32, 59]. We prepare two samples for each concentration. For each sample, we thoroughly mixed it using a stirring rod and then allowed it to stand to eliminate foam. Each prepared sample was stored in a standard 20mL plastic sample bottle, filled to capacity.

Flash-only image collection. To capture flash-only images of liquid samples, we acquire paired images under both flash-illuminated and ambient-only lighting. The flash-only

Table 3: Accuracy of the Spectral Recovery Model.

Model	MRAE	RMSE
SOTA	0.1385	0.1049
RVN-SR w/o ECDA	0.0893	0.0603
RVN-SR with ECDA (ours)	0.0751	0.0424

component is isolated by subtracting the ambient-only image from the flash-illuminated image, with consistent camera settings (shutter speed, ISO, focus, white balance) maintained throughout using a professional photography app. In default, data are collected in an artificially lit conference room with the iPhone 11 handheld approximately 10 cm above the liquid surface. The system is as simple to use as a standard smartphone camera: focus, then shoot, without needing additional stabilization. For each sample, we gain four flash-only images: two with an open container and two sealed. We have (34 honey sample categories + 22 wine sample categories) $\times$ 2 samples $\times$ 4 captures = 448 images under default experimental setting.

ROI extraction. We manually select liquid surface regions of interest (ROI, 20 $\times$ 20 pixels) in honey samples for spectral recovery. This sampling strategy produce 400 RGB values per flash-only image. Here, each pixel is treated as an individual sample for classification. Thus, our dataset comprised 179,200 pixel-level samples extracted from 400-pixel ROIs across all 448 captures.

Illumination intensity normalization. Due to variations in camera distance during manual data collection, brightness inconsistencies arise. To address this, we perform per-pixel brightness normalization: for each pixel, the maximum RGB value is scaled to the full intensity range, and the other two channels are adjusted proportionally. This preserves color ratios while ensuring uniform brightness. The normalized pixels are then used for further analysis.

We then feed the preprocessed RGB data into the trained spectral recovery model to infer the VIS-NIR spectra. These inferred spectra are subsequently used to train the liquid adulteration detection model.

7.3 Accuracy of the Recovered Bands

In this section, we evaluate the performance of our spectral recovery model (Section 6), validating its effectiveness as the core component of LiquidLens. We first assess the core spectral recovery performance on known liquids, and then extend the evaluation to unseen liquids and adulterants to test generalization.

7.3.1 Known Liquid and Adulterant. This evaluation is conducted on the two datasets described in Section 7.1, which contains paired RGB data and corresponding ground-truth

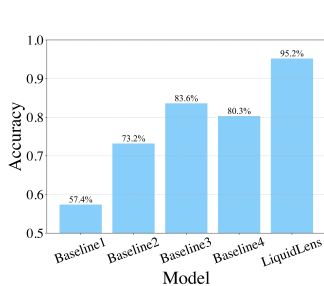


Figure 14: Comparison between LiquidLens and baseline models.

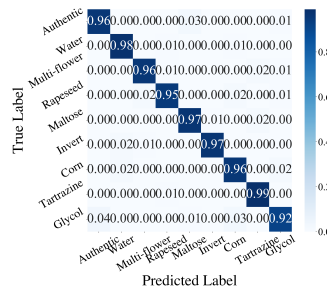


Figure 15: Accuracy of honey adulteration detection.

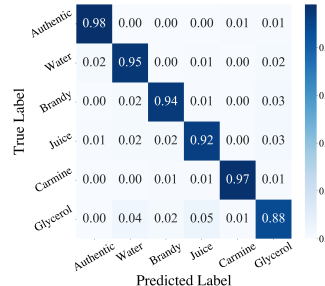


Figure 16: Accuracy of wine adulteration detection.

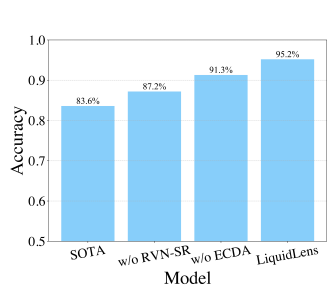


Figure 17: Ablation Study.

hyperspectral data. Note that we use both datasets to train the model together, aiming to recover full-band VIS-NIR hyperspectral data from flash-based RGB data. We divide the data of the two datasets into two disjoint partitions: 70% for training and 30% for testing. We test the accuracy of the recovered bands by comparing them against the ground truth ones. With an inter-band distance of only 5 nm, the recovered spectra exhibit high spectral resolution, sufficient for detecting liquid adulteration. Accuracy is measured using standard metrics: Mean Relative Absolute Error (MRAE) and Root Mean Square Error (RMSE).

We employ SOTA-reconstructed VIS-NIR data (Section 6.1.1) as our baseline, with output dimensions extended for NIR recovery. Two variants of our spectral recovery model are evaluated: without ECDA and with ECDA (our full design). Results in Table 3 show metrics close to zero, indicating high spectral similarity between recovered and ground-truth bands. Moreover, our model outperforms both the SOTA baseline and the non-ECDA variant, confirming its effectiveness.

7.3.2 Unknown Liquid and Adulterant. To validate the generalization of our spectral recovery model to unseen liquids and adulterants, we conduct an additional experiment using orange juice. This validation uses only hyperspectral data (under full-band illumination) to assess spectral reconstruction accuracy, and is independent of the mobile RGB dataset (Section 7.2 and Table 2) used for primary detection. We evaluate hyperspectral reconstruction accuracy for orange juice adulterated with water, syrup, and tartrazine. Ten samples are prepared: two authentic (NFC and reconstituted) and three adulterated NFC (with 40% water, 40% syrup, or 0.2 g/kg tartrazine), each in duplicate. A single hyperspectral image is captured for each sample. From each image, 100 pixels are selected from the ROI, yielding a final dataset of 1,000 spectral curves (10 samples $\times$ 100 pixels each).

Table 4: Accuracy of the Spectral Recovery Model for Unknown Liquid and Adulterant.

Model	MRAE	RMSE
New liquid w/o transfer learning	0.3660	0.3516
New liquid with transfer learning	0.0458	0.0397
New adulterant w/o transfer learning	0.0621	0.0534

Table 5: Specular Reflection Removal Result.

	Flash&Ambient	Ambient-only	Flash-only
Exp.			
Var.	2.61	3.78	1.24

We adopt a transfer learning strategy: freezing all layers of the pre-trained spectral reconstruction and NIR prediction models, and fine-tuning only the final output layers. This preserves learned feature representations while efficiently adapting to new data. With only 20 randomly selected spectral curves (out of 1,000) for fine-tuning, this approach yields substantial performance gains (Table 4), demonstrating that our framework can achieve high accuracy on new liquid types with minimal data.

We further tested the model on unknown adulterants by training only on authentic orange juice spectra and evaluating on adulterated samples. Results (Table 4) show minimal accuracy degradation, demonstrating robustness. This is because common adulterants (e.g., water, syrup) share key chemical bonds (O-H, C-H) with the authentic juice, differing mainly in concentration. Their low dosage places the adulterated samples close to the authentic ones in the feature space, allowing the model to approximate them effectively.

7.4 Results of Flash-only Specular Reflection Removal

We evaluate the specular reflection removal effect of flash-only method in Section 5. Traditional approaches in computer vision always employ PSNR and SSIM [36] to evaluate reflection removal effects. However, calculating the two metrics requires the groundtruth images without specular reflection, which can not be acquired in our scenario.

We employ spatial-frequency analysis to evaluate reflection removal. Specular reflections appear as high-frequency components, while authentic liquid textures occupy lower bands. Using patch-based variance (computing local RGB intensity variance across sliding windows), we analyze three image types: flash-illuminated, ambient-only, and their difference (flash-only). We test 50 sets of the three image types. Table 5 shows example images and average variance. Results confirm flash-only images exhibit substantially reduced high-frequency energy, validating our method’s effectiveness in specular removal.

7.5 Accuracy of Liquid Food Adulteration Detection

In this section we evaluate the performance of LiquidLens in detecting types of liquid food adulteration across various experimental settings. By default, we report the average performance from an 8-fold cross-validation, ensuring that the class ratio in each fold remains consistent with that of the entire dataset. Note that pixels from the same image are never split across the training and test sets simultaneously.

7.5.1 Baselines Comparison. We compare our system’s performance with several baselines. (1) Ambient-only RGB. We use the RGB data without flash light illuminated to conduct the classification. (2) Flash-only RGB. We use flash-only RGB data to conduct the classification. (3) Hyperspectral data in the visible range reconstructed from flash-only RGB images using SOTA methods. We employ the SOTA spectral reconstruction method (detailed in Section 6.1.1) to generate hyperspectral data in the visible range from flash-only RGB inputs. (4) VIS-NIR hyperspectral data reconstructed by adapting our SOTA model with extended output dimensions for NIR spectral recovery.

Fig. 14 compares the accuracy of the LiquidLens with baselines, revealing three important findings. 1) Using RGB for liquid adulteration detection results in poor performance, which is lower than 80% accuracy, due to the limited spectral range and low spectral resolution. 2) Detection performance using ambient-only RGB is significantly inferior to flash-only RGB, primarily due to interference from variable ambient illuminance and liquid specular reflections. 3) The

existing SOTA method improves system performance by reconstructing the visible spectrum from flash-only RGB data. However, due to its failure in NIR spectrum recovery and substantial domain shift, the detection accuracy remains below 85%. 4) The existing SOTA method demonstrates lower detection accuracy when reconstructing VIS-NIR spectra compared to visible spectrum reconstruction, primarily due to inaccuracies in NIR reconstruction that introduce noise interference in liquid adulteration detection. Our designed system achieves 95.2% accuracy for the liquid adulteration type detection task, outperforming all baseline methods and validating the effectiveness of our system design.

7.5.2 LiquidLens’s Performance. Fig. 15 and Fig. 16 presents the confusion matrix of LiquidLens for detecting honey and wine adulteration types. The results demonstrate consistently good performance across different liquid food categories and adulteration types. LiquidLens effectively distinguishes between various types of adulterated samples, achieving an overall accuracy of 96.2% for honey and 94.0% for wine.

7.5.3 Ablation Study. Ablation studies quantify each module’s contribution (Fig. 17). Replacing our RVN-SR module with a SOTA baseline causes an 8.0% accuracy drop, underscoring its importance. RVN-SR employs a sparse-gated transformer to enable precise NIR analysis from visible spectra. Removing the ECDA framework (i.e., training RVN-SR end-to-end without domain adaptation) results in a 3.9% accuracy drop, confirming ECDA’s role in bridging the illumination domain gap and compensating for missing NIR data.

7.5.4 Different environments. We evaluated authentic and adulterated manuka samples in five environments, which varied in spectral composition, intensity, and surrounding reflection characteristics. The test environment’s data are excluded from the training set. These environments are carefully selected to represent common real-world scenarios: 1) In a near-dark environment (10 lux) with minimal ambient illumination, creating a simple reflection scenario. 2) In a conference room with the lighting infrastructure on at night (200 lux), creating a simple reflection scenario. 3) In a daytime home setting (500 lux), creating a complex reflection environment with various surfaces including walls and equipment. 4) In a daytime office with lighting infrastructures on (800 lux), creating complex light interactions with walls and equipment surfaces. 5) In a daytime office besides a window (1500 lux), featuring a simple reflective environment.

Fig. 18 presents the adulteration detection accuracy under the different conditions. We can observe that 1) under common indoor lighting scenarios (≤ 800 lux), the system

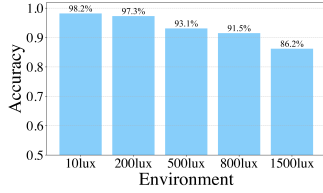


Figure 18: Accuracy under different environments.

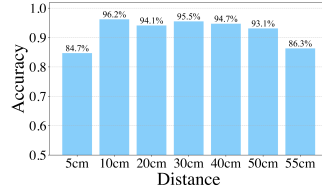


Figure 19: Accuracy under different distances.

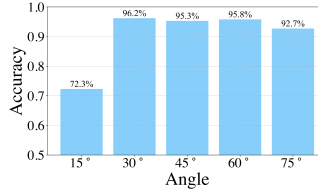


Figure 20: Accuracy under different angles.

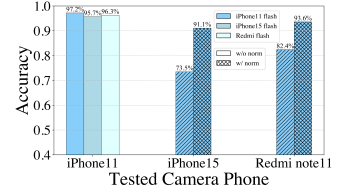


Figure 21: Accuracy on different devices.

achieves the accuracy ranging from 91% to 99%, which confirms that LiquidLens can work in diverse real-world environments. 2) Under extremely high ambient illumination, detection accuracy decreases significantly. This occurs because under the fixed camera parameters of our experimental setup, intense ambient light can drive the sensor signal toward saturation. This saturation corrupts the differential imaging process, compromising the extraction of the isolated flash component. 3) With more complex reflection scenario, the detection accuracy becomes lower. The reason is that although flash-only imaging eliminates dominant specular and ambient noise, residual artifacts from multi-path propagation and complex surface interactions under complex reflection scenario may remain.

7.5.5 Different distances. We analyze the impact of the capturing distance on the accuracy of the liquid adulteration detection by LiquidLens. We capture the authentic and adulterated manuka samples at 8 distances: 5 cm, 7 cm, 10 cm, 20 cm, 30 cm, 40 cm, 50 cm, 55 cm. The test distance’s data are excluded from the training set. As shown in Fig. 19, we have three observations: 1) A significant performance degradation occurs at the 5 cm distance. The reason is that lens defocusing at extremely close distances can generate chromatic aberrations in the measured spectra [2]. 2) Distance variations between 7-50 cm demonstrate consistent detection accuracy, indicating robust distance generalization of the LiquidLens algorithm. The reason is that we apply brightness normalization to eliminate interference caused by variations in the distance between the camera flash and the target liquid. 3) Performance declines significantly at 55 cm. The reason is that the camera flash becomes too weak to provide adequate illumination at long distance, causing low signal-to-noise ratio that distorts color accuracy.

7.5.6 Different angles. We quantify the impact of the camera’s tilt angle on system performance. We capture images of authentic and adulterated Manuka honey samples with the camera positioned at a fixed distance of 10 cm and tilted at angles of 15°, 30°, 45°, 60°, and 75° relative to the surface normal. Results in Fig. 20 show robust performance from

30° to 75°. Although the reflected intensity changes with angle (according to the lambert radiation model [39, 40]), our brightness normalization successfully removes this interference, allowing the essential color features to be retained for analysis. The system fails at the angle below 15° because the specular reflection from the flash falls within the camera’s field of view (FOV). At larger angles, this reflection is steered outside the camera’s FOV, allowing only the diffuse signal to be captured. Note that a 90° angle is not feasible as the liquid surface falls entirely outside the camera’s field of view in that configuration and is therefore excluded from evaluation.

7.5.7 Different devices. We capture authentic and adulterated manuka samples with 3 different devices, iPhone 11, iPhone 15, Redmi note 11. The model was trained on a dataset collected using the iPhone 11’s camera and flashlight.

Camera flashes on different devices. First, we evaluate LiquidLens under illumination from camera flashes on various mobile devices. Specifically, we use the camera flashes of iPhone 11, iPhone 15 and Redmi note 11 as light sources separately, while employing an iPhone 11 to capture the corresponding RGB images. Results in Fig. 21 demonstrate that LiquidLens performs robustly under different camera flashes, due to the consistent spectral characteristics exhibited across these light sources.

Cameras on different devices. We then evaluate LiquidLens using the same iPhone 11 flash as the light source but with different cameras (i.e. iPhone 11, iPhone 15, and Redmi Note 11) to capture RGB images. The experimental results in Fig. 21 reveal significant performance degradation due to the variations in spectral response characteristics across different camera sensors. To mitigate this cross-device discrepancy, we adopt the image normalization technique proposed in [55], which requires only a small calibration dataset per device. We trained a deep learning encoder-decoder architecture to project all input images into a unified color space aligned with the iPhone 11’s spectral characteristics.

7.6 Time Consumption and Memory Usage

We test the time consumption and the memory usage on two mobile devices: Redmi note 11 and OnePlus Ace5. The

average processing time of LiquidLens for each pixel is 0.21 msec on Redmi note 11 and 0.14 msec on OnePlus Ace5. For each device, we test 400 pixels and calculate the average time. LiquidLens used less than 90 MB of memory on both devices.

8 Related Work

Spectral-based Liquid Food Adulteration Detection. Spectral analysis has become a powerful tool for detecting liquid food adulteration. Based on wavelength ranges, it can be categorized into three types: visible, VIS-NIR, and NIR. Visible spectral analysis (400–700 nm) is primarily used to identify colour-based adulterants such as artificial dyes in liquids [62], though it lacks the sensitivity to distinguish chemically similar substances. VIS-NIR analysis (400–1000 nm) offers higher sensitivity, enabling the detection of more subtle adulterations such as sugar syrup added to honey [6, 54]. NIR analysis (900–1700 nm) provides even greater sensitivity and is widely applied in detecting chemical adulterants such as melamine in milk [12, 17, 43] and assessing quality indicators in olive oil [21, 45]. However, these methodologies require bulky and expensive hyperspectral devices.

Liquid Analysis on Mobile Devices. Previous mobile liquid analysis methods [30, 38, 70, 74] primarily rely on measuring physical properties, including viscosity and surface tension. However, their measurements need direct contact with container, thus are easily compromised by interference from the container material due to resonant effects. [11] sense the viscosity from a drop of liquid, which can not be interfered by container material. Nevertheless, the requirement for controlled deposition of liquid droplets onto glass slides adds significant procedural overhead. LiquidHash[60] leverages smartphone cameras to record videos of bubble dynamics (including morphology and motion patterns) to detect liquid food adulteration. However, the system requires a specialized bottle cap accessory to regulate bubble formation for reliable measurements.

Hyperspectral Sensing on Mobile Devices. While most existing mobile sensing systems rely on wireless sensing techniques [23, 42, 71, 72, 76] to capture physical properties of objects, mobile hyperspectral sensing emerges as a new paradigm for chemical analysis. Recent works, e.g., [22, 33, 58, 63, 67], have explored the implementation of hyperspectral sensing on mobile devices. However, most of these approaches require hardware modifications. For instance, Lumos [67] integrates a spectral sensor and an LED array into a wristband to enable hyperspectral capture. ChromaFlash [63] employs a specialized multispectral flash unit attached to a smartphone to reconstruct hyperspectral data in the visible range using the device’s camera. Several works [27, 55] achieve hyperspectral reconstruction across VIS-NIR bands using only smartphone cameras. However, it relies on

a NIR camera module, which is not commonly available in most commercial smartphones.

9 Discussion

Outdoor Deployment. While our experimental setup uses fixed camera parameters (such as ISO and shutter speed) optimized for indoor environments, these settings may not be suitable for outdoor use. Strong ambient light outdoors can cause image saturation, reducing the effectiveness of our method. Future work should address this limitation by calibrating camera parameters for outdoor conditions and dynamically adjusting exposure to ensure robust bulk scattering extraction under varying lighting environments.

Target Liquid Requirements. Our method’s effectiveness depends on two key conditions. First, it requires distinct visible absorption features to reconstruct the NIR component; therefore, high accuracy is achieved mainly for colored liquids, as transparent ones lack sufficient spectral variation in RGB signals. Second, it assumes liquid homogeneity, which we ensure by stirring, so the spectral signature from the surface represents the entire volume; without homogeneity, the analysis remains local to the imaged layer.

From Single Pixel to Full ROI. Our study validates single-pixel spectral recovery for liquid adulteration detection, relying on the sample homogeneity assumption whereby any pixel’s spectrum represents the whole sample. We segment a liquid ROI and apply the model to one pixel within it. In practice, this framework supports flexible extension: detection can be performed either on the averaged color of the entire ROI, or on each pixel individually with results consolidated by voting.

10 Conclusion

LiquidLens enables mobile VIS-NIR spectral analysis for liquid food adulteration detection using only the RGB sensing capability. By utilizing the camera flash as a spectrally consistent light source and innovating with flash-only reflection suppression and sparsity-optimized spectral recovery, we overcome fundamental limitations in uncontrolled environments and spectral sensing range of mobile devices. Experimental validation demonstrates robust performance across honey (96.2% accuracy) and wine (94.0% accuracy) adulteration detection under diverse real-world conditions.

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